KANG Extended Roadmap v5.2

v5.1 + review changes (2026-10-31). Ye v5.0 ko replace nahi karta.

1. v5.1 se kya badla

1. Phase 2 (Semantic Search) aur Phase 6 (Smart Matching) **merge**: dono CLIP/embedding wale kaam hain. Order ab Ingest → rule-based Matcher → CLIP.
2. **Hindi queries:** original CLIP sirf English samajhta hai. Multilingual model (M-CLIP / multilingual SigLIP) ya query-translate step zaroori. Tab tak alias table (**मंदिर**→temple, **नागेश्वर**→nageshwar…).
3. **Vector DB baad mein:** 10k assets tak numpy cosine. FAISS/Chroma/Qdrant sirf jab library 10k+ ho jaye.
4. **Ek renderer:** FFmpeg canonical renderer hai (abhi PASS). MoviePy ya Editly mein se sirf ek evaluate hoga (Editly = Node.js runtime extra).
5. **Premiere ke liye FCP7 XML writer hi rahega** (H-layout, 24.6.1 par PASS). OTIO sirf internal/neutral model, DaVinci jaise editors ke liye baad mein.
6. **Commands vendor-neutral:** har /kang-* = `kang.py <subcommand>`. Claude skills/plugins sirf thin wrapper; Antigravity ya koi aur agent same CLI chala sake.
7. **Baseline = v5_1 (KANG_008)**: CREDITS.txt/json package ke andar aur manifest hash mein, `visual_track` flag, relative assets_dir. Premiere evidence aate hi manifest update.
8. **Scope gate:** Shorts, talking-head, multi-format (Phase 5, 7) M3 exit ke baad: Dwarika text, sync aur rights close hone par (roadmap v2 rule).

2. Architecture (same as v5.1)

Layer	Kaam	Policy
Core Engine	FFmpeg, ASR, asset index, timeline, renderer, QA, manifest	Stable, versioned
KANG Skills	kang.py subcommands: ingest, match, approve, build, premiere	Add/remove
Plugins	Claude skills, MCP servers, editor bridges	Optional wrappers
Profiles	Travel doc, talking-head, Shorts/ Reels, faceless	Rules per format
Projects	`runs/<date>_KANG_NNN`, immutable inputs, manifest	Never overwrite

3. Implementation plan (revised)

Phase	Deliverable	Success criteria	Status
0 Baseline lock	v5_1 package + evidence	Premiere screenshot/ evidence in manifest	Premiere test chal raha hai
1 Asset Ingest	`kang.py ingest` → index.json, contact sheet	54 Dwarika assets indexed, licence per asset	**Done (2026-10-31)**, 4 tests PASS
2 Rule Matcher	`kang.py match` → broll_suggestions.json	Har scene ke 3 candidates + reason; user approve	Next
3 Semantic (CLIP)	multilingual embeddings, numpy cosine	Untagged library par test queries ka top-3 relevant	Evaluation
4 Skills wrap			Proposed

Phase	Deliverable	Success criteria	Status
	/kang-new, /kang-ingest, /kang-broll, /kang-build	Ek command se setup → preview	
5 Multi-format	Shorts/Reels adapter, OTIO internal	Ek project se long-form + Shorts	M3 exit ke baad
6 Talking-head	silence/filler cuts (auto-editor evaluate), retakes	Raw take se clean rough cut	M3 exit ke baad
7 Plugin ecosystem	MCP servers, community plugins	Nayi capability bina core rewrite	Future

4. Tools: decision rule same (isolated test → licence/security → adopt/adapt/reject)

Tool	Phase	Note
Multilingual CLIP / SigLIP	3	GTX 1650 (4 GB) par ViT-B/32 size theek
MoviePy ya Editly	4	Sirf ek; FFmpeg canonical
ffmpeg-skill	reference	Hamara FFmpeg layer already hai; ideas lo
OTIO	5	Internal only; Premiere = FCP7 XML
auto-editor, openshorts, auto-subs, FunClip	5–6	P2 evaluation
OpenCut-AI	3	Matching reference only; cloud/paid exclude
Concat, pyJianYingDraft	—	Jab tak CapCut output na chahiye
LivePortrait, VideoReTalking, Hypit	—	Reject for now

5. Data contracts

- **assets/index.json (kang.assets/1.0, implemented):** asset_id (sha256[:12]), path, type, width/height/duration, sha256, tags, description, title, author, source_url, licence, licence_url, licence_info (family, sharealike, attribution_required, commercial_ok), embedding_ref (null until Phase 3), rights_status (recorded_not_cleared / unknown).
- **broll_suggestions.json (Phase 2):** scene_id, candidates [asset_id, score, reason], approval_status.
- visuals.json, qa_report.json, manifest.json: v5.1 jaisa. schema_version/config_hash/tool_versions abhi planned.

6. Honest note on Phase 1 test

Dwarika ke tags file names aur briefs se bane hain, jo selection ke waqt hi likhe gaye the. Isliye "**नागेश्वर मंदिर** → S13" jaisi search ka sahi aana **expected hai, proof nahi**. Asli matching test Phase 3 mein bina-tag wali library par hoga.

7. Rights (same rule)

54 assets: 48 CC BY-SA, 3 CC BY, 2 PD, 1 CC0. Clearance alag gate hai. Options: SA assets ko contain-fit, ya CC BY/CC0/PD/apni photos se swap. `asset-search --no-sharealike` swap candidates dikhata hai. Ye legal advice nahi hai.